

FreqSpatial: Plug-and-Play Frequency-Spatial Feature Enhancement for Real-Time DETR Detection

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Abstract—Real-time DETR-based detectors achieve competitive speed-accuracy trade-offs, yet their feature pyramid network (FPN) necks rely exclusively on spatial-domain convolutions, overlooking frequency-domain information that captures fine-grained textures and global structural patterns. We propose FreqSpatial, a dual-branch block that fuses Scharr-based edge enhancement with FFT-based spectral convolution inside a drop-in CSP-style module. Replacing the standard CSPRepLayer in RT-DETRv2’s neck with FreqSpatial yields +1.45 mAP on COCO 2017 (46.73% $\rightarrow$ 48.18%) without altering the backbone or decoder. The 72-epoch FreqSpatial model matches the official 120-epoch RT-DETRv2-S performance, indicating that frequency-aware inductive bias accelerates convergence. A +3.4 mAP margin at epoch 0 further supports this conclusion. FreqSpatial operates as a drop-in CSP block; the underlying frequency-spatial module is architecture-agnostic and may generalise to other FPN-style necks. Our implementation is open-source.

I. INTRODUCTION

Object detection has advanced considerably with the emergence of DETR-based architectures [1], which formulate detection as a set prediction problem and eliminate hand-crafted components such as non-maximum suppression and anchor generation. RT-DETR [2] further advances this paradigm with a real-time hybrid encoder-decoder that achieves competitive speed-accuracy trade-offs. However, the FPN neck in RT-DETR relies exclusively on spatial-domain convolutions for multi-scale feature fusion, overlooking frequency-domain information that encodes fine-grained textures and structural patterns not easily captured in the spatial domain.

Frequency-domain features represent global structural information independent of spatial position. The Fourier transform decomposes spatial signals into orthogonal frequency components, enabling a model to capture periodic patterns and fine details that the spatial domain may attenuate. FSDETR [12] recently proposed frequency-spatial fusion for DETR-based detection and achieved strong results on small-object benchmarks (VisDrone, TinyPerson). FD-RTDETR [14] explored frequency enhancement inside RT-DETR’s encoder but reported modest gains (+0.7 mAP on COCO). The potential of frequency-spatial fusion that combines Scharr-based edge extraction with FFT convolution remains underexplored on large-scale general-purpose detection.

We propose **FreqSpatial**, a frequency-spatial feature enhancement block for the FPN neck of RT-DETR. Building on

the Scharr edge enhancement and FFT spectral convolution introduced in FSDETR [12], we redesign the module as a CSP-style drop-in block and conduct a large-scale validation on COCO 2017. FreqSpatial uses a dual-branch architecture: a spatial branch with Scharr operators for local gradient enhancement, and a frequency branch that performs convolution in the Fourier domain to capture global spectral patterns. The two branches fuse within a CSP-style residual structure, directly replacing standard CSPRepLayer blocks.

Experiments on COCO 2017 show that the RT-DETRv2-S (PResNet18) baseline reaches 46.73% mAP, while FreqSpatial attains **48.18% mAP** (+1.45). At epoch 0, before any detection training, FreqSpatial achieves 17.8% mAP vs. 14.4% for the baseline (+3.4), indicating that frequency-aware inductive bias provides an immediate advantage.

Our contributions are:

- 1) We validate frequency-spatial feature fusion at scale on COCO (80 classes, 118K images), showing that FreqSpatial generalizes beyond small-object benchmarks to general-purpose detection (+1.45 mAP).
- 2) We redesign the frequency-spatial module as a CSP drop-in block that integrates into RT-DETR’s FPN+PAN neck without backbone or decoder changes.
- 3) We provide efficiency analysis, cold-start experiments, ablation against attention-based alternatives, and qualitative visualizations, with an open-source implementation.

II. RELATED WORK

A. Real-Time DETR-based Detectors

DETR [1] reformulated object detection as a set prediction problem using transformers. Deformable DETR [3] replaced dense attention with deformable attention to improve efficiency. DINO [4] introduced contrastive denoising and mixed query selection. RT-DETR [2] proposed a hybrid encoder combining intra-scale interaction with cross-scale fusion for real-time inference. RT-DETRv2 [13] refined the encoder with improved training recipes and a set of freebies for real-time detection. Our work enhances RT-DETRv2’s neck with frequency-domain features via a drop-in CSP block, complementing its existing encoder design.

B. Frequency-Domain Feature Learning

The Fourier transform captures global frequency characteristics in vision. FcaNet [9] introduced frequency-domain

channel attention. GFNet [10] employed learnable global frequency filters for ViTs. FSDETR [12] introduced frequency-spatial awareness for small-object detection on VisDrone and TinyPerson, combining Scharr operators with FFT-based convolution. However, FSDETR focused on specialized benchmarks with limited class counts (1–10) and did not evaluate on general-purpose detection. FD-RTDETR [14] applied frequency enhancement to RT-DETR’s intra-scale feature interaction, attaining +0.7 mAP on COCO and demonstrating the potential of frequency-domain processing for general detection. Our work adopts FSDETR’s Scharr+FFT design and repackages it as a CSP neck block for general-purpose detection at scale—80 classes, 118K images on COCO 2017, achieving +1.45 mAP. The gain doubles that of FD-RTDETR (+0.7 mAP), which applies frequency enhancement to RT-DETR’s intra-scale encoder rather than the FPN neck.

C. Feature Pyramid Networks

FPN [5], PANet [6], and BiFPN [7] are standard architectures for multi-scale feature fusion. CSPNet [8] proposed cross-stage partial connections to reduce computation while preserving feature quality. We integrate frequency-spatial processing into CSP-style fusion blocks, applied across both FPN and PAN pathways.

III. METHOD

A. Preliminaries: RT-DETR Hybrid Encoder

RT-DETR’s hybrid encoder has two components: (1) an intra-scale transformer encoder on the top-level feature map (stride-32), and (2) a cross-scale FPN+PAN fusion path. Given backbone features $\{C_3, C_4, C_5\}$ at strides $\{8, 16, 32\}$, each is projected to $d = 256$ via 1×1 convolutions, then processed through:

- **Top-down FPN:** Higher-level features are upsampled and fused with lower-level features via CSPRepLayer blocks.
- **Bottom-up PAN:** Lower-level features are downsampled (stride-2 3×3 conv) and fused with higher-level features via CSPRepLayer blocks.

Each fusion block takes $F_{in} \in \mathbb{R}^{B \times 2d \times H \times W}$ (concatenated upsample+lower or downsample+higher) and outputs $F_{out} \in \mathbb{R}^{B \times d \times H \times W}$.

B. FreqSpatial Block

We replace CSPRepLayer with FreqSpatialBlock in both FPN and PAN paths. It inherits the CSP two-path architecture but uses frequency-spatial bottlenecks. Given input F_{in} :

$$F_{out} = \text{Conv}_{1 \times 1}^{out} \left(\underbrace{\text{Conv}_{1 \times 1}^a(F_{in})}_{\text{path a}} + \underbrace{\text{Conv}_{1 \times 1}^b(\mathcal{F}^N(\text{Conv}_{1 \times 1}^a(F_{in})))}_{\text{path b}} \right) \quad (1)$$

where $\mathcal{F}$ is the FreqSpatial bottleneck applied $N = 3$ times.

C. FreqSpatial Module

As shown in Fig. 1, FreqSpatial has two parallel branches, taking an input tensor $X^{\text{Input}} \in \mathbb{R}^{B \times C \times H \times W}$.

1) *Spatial Branch: Scharr Edge Enhancement:* Fixed Scharr operators extract edge gradients with better rotational symmetry than Sobel:

$$\mathbf{G}_x = \begin{bmatrix} 3 & 0 & -3 \\ 10 & 0 & -10 \\ 3 & 0 & -3 \end{bmatrix} * X, \quad \mathbf{G}_y = \begin{bmatrix} 3 & 10 & 3 \\ 0 & 0 & 0 \\ -3 & -10 & -3 \end{bmatrix} * X \quad (2)$$

Edge magnitude $\frac{1}{2}(\mathbf{G}_x + \mathbf{G}_y)$ is refined through two 3×3 convs with a residual connection:

$$F_{\text{spatial}} = \text{Conv}_{3 \times 3}^2(\text{Conv}_{3 \times 3}^1(\text{Scharr}(X)) + X) \quad (3)$$

2) *Frequency Branch: FFT Spectral Processing:* The input is transformed to the Fourier domain:

$$\mathcal{F}(X) = \text{rFFT2D}(X) \in \mathbb{C}^{B \times C \times H \times (\lfloor W/2 \rfloor + 1)} \quad (4)$$

Real and imaginary parts are separated, concatenated along channels, and processed by a 3×3 convolution in the frequency domain:

$$F_{\text{freq}} = \text{Conv}_{3 \times 3}(\text{Concat}[\Re(\mathcal{F}(X)), \Im(\mathcal{F}(X))]) \quad (5)$$

The processed features are reassembled into complex form and inverted:

$$F_{\text{freq}}^{\text{spatial}} = \mathcal{F}^{-1}(\text{Complex}(F_{\text{freq}})) \quad (6)$$

followed by a 3×3 refinement convolution.

3) *Fusion:* The two branch outputs are element-wise summed, then fused through a 1×1 convolution to produce the final enhanced features X^{Output} :

$$F_{out} = \text{Conv}_{1 \times 1}(F_{\text{spatial}} + F_{\text{freq}}^{\text{spatial}}) \quad (7)$$

D. Design Rationale

The two branches provide complementary information: the Scharr branch captures local edge structures and high-frequency spatial details, while the FFT branch encodes global periodic patterns and low-to-mid frequency textures. FreqSpatialBlock preserves the same input-output interface as CSPRepLayer—a true drop-in replacement requiring no backbone or decoder changes. The main cost is increased FLOPs from FFT operations and additional convolutions (Section IV-E).

IV. EXPERIMENTS

A. Experimental Setup

We use COCO 2017 [15] (118K train / 5K val images, 80 classes). All models use RT-DETRv2-S with PResNet18 backbone (ImageNet pretrained) and hidden dimension $d = 256$. Training: AdamW optimizer (base learning rate 10^{-4} , backbone learning rate 10^{-5}), linear warmup (2000 iterations), AMP, batch size 16, input size 640×640 , 72 epochs on one NVIDIA RTX 3090. Metrics: mAP, AP₅₀, AP₇₅, AP_S, AP_M, AP_L, AR^{max=100}.

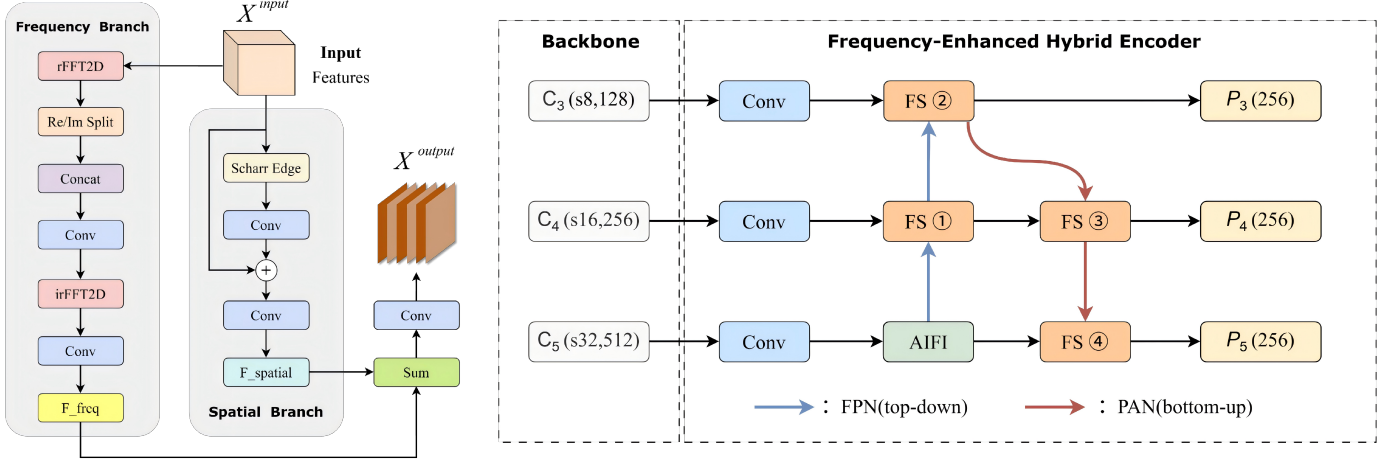


Fig. 1: Overview of the proposed method. (a) Architecture of the FreqSpatial module: a spatial branch using Scharr edge detection with two 3×3 convolutions and a residual (+) connection, and a frequency branch processing features through $rFFT2D$, split, convolution, and $irFFT2D$; outputs are summed and fused via 1×1 convolution. (b) The RT-DETR hybrid encoder: backbone features C_3 – C_5 are projected to 256 channels, then processed through AIFI (green) and four FreqSpatial (FS) Blocks (orange) in the FPN and PAN paths. FS Blocks replace standard CSPRepLayer; backbone, AIFI, and decoder unchanged.

TABLE I: Detection performance on COCO 2017 val. All models based on RT-DETRv2-S (PResNet18), trained 72 epochs. Δ denotes improvement of FreqSpatial over Baseline.

Method	mAP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L	AR ¹⁰⁰
Official (120ep)	48.1	–	–	–	–	–	–
Baseline	46.73	63.6	50.6	29.8	49.9	62.7	69.5
FreqSpatial	48.18	65.2	52.1	30.8	51.5	64.2	70.3
Δ	+1.45	+1.6	+1.5	+1.0	+1.6	+1.5	+0.8

B. Main Results

Table I reports COCO 2017 validation results. FreqSpatial achieves **48.18% mAP**, a +1.45 improvement over the Baseline (46.73%). Gains were consistent across scales: +1.6 AP₅₀, +1.5 AP₇₅, +1.6 AP_M, +1.5 AP_L, and +1.0 AP_S. The gain on small objects was smaller than on medium and large objects. This pattern is consistent with the cold-start results (Table II), where large objects benefit most from frequency-domain features, and suggests that global spectral patterns are inherently less informative for small instances. At 72 epochs, FreqSpatial matches the official 120-epoch RT-DETRv2-S performance (48.1% [13]), reaching comparable accuracy with 40% fewer training epochs.

C. Cold-Start Analysis

Table II reports epoch 0 validation, before detection training (ImageNet pretrained backbone only). FreqSpatial achieves 17.8% mAP vs. 14.4% for the Baseline (**+3.4 mAP**). The gain was largest on large objects (+6.2 AP_L), where global frequency patterns are most informative.

TABLE II: Epoch 0 validation (no detection training).

Method	mAP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
Baseline	14.4	21.8	15.6	9.7	17.9	18.3
FCM	13.9	21.4	14.9	8.6	16.4	18.3
FreqSpatial	17.8	26.3	18.9	9.8	20.7	24.5
Δ (vs Base)	+3.4	+4.5	+3.3	+0.1	+2.8	+6.2

TABLE III: COCO mAP progression during training.

Method	Ep 0	Ep 50	Ep 59	Ep 67	Ep 71
Baseline	14.4	45.8	46.1	46.4	46.73
FreqSpatial	17.8	47.49	48.17	48.17	48.18

D. Training Dynamics

Table III and Fig. 2(a) track mAP during training. FreqSpatial reaches 47.49% by epoch 50, already exceeding the final Baseline (46.73%). mAP plateaus at 48.17% from epoch 59 onward, with the best result of 48.18% at epoch 71. This trajectory indicates faster convergence associated with frequency-aware features. Further gains under the current learning rate schedule appear limited.

E. Efficiency Analysis

Table IV and Fig. 2(b) compare computational cost. FreqSpatial increases parameters from 20.18M to 69.03M and FLOPs from 30.38G to 118.43G, driven primarily by FFT operations and extra convolutions. FPS decreases from 66.4 to 38.5 on an RTX 3090 ($1.7 \times$ reduction). At 38.5 FPS, inference remains above the conventional real-time threshold of 30 FPS.

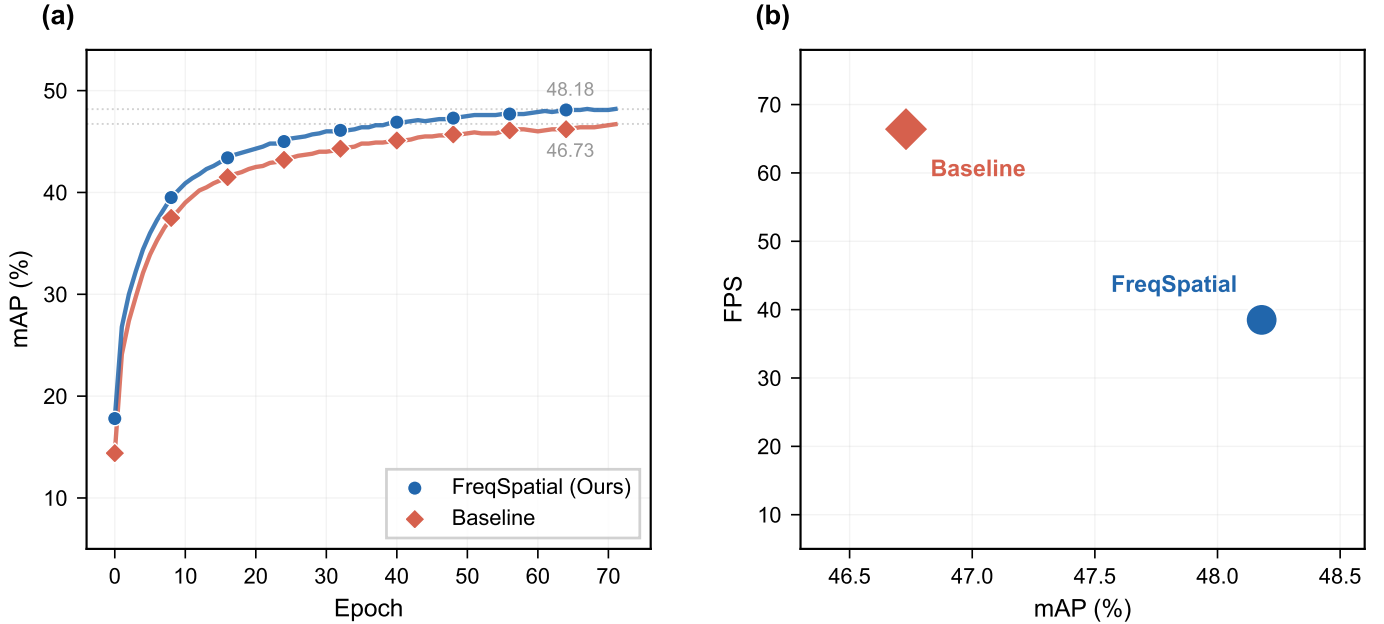


Fig. 2: Training dynamics and efficiency analysis. (a) mAP progression during 72-epoch training. FreqSpatial converges faster and surpasses the final Baseline (46.73%) by epoch 50; dotted horizontal lines mark final mAP of each model. (b) Accuracy-speed trade-off on an RTX 3090. FreqSpatial achieves higher mAP (48.18%) at 38.5 FPS, remaining well above the real-time threshold.

TABLE IV: Efficiency on RTX 3090 (input 640×640 , batch=1).

Method	Params (M)	GFLOPs	FPS	Latency (ms)
Baseline (CSPRepLayer)	20.18	30.38	66.4	15.1
FCM	20.33	30.76	48.2	20.8
FreqSpatial	69.03	118.43	38.5	26.0

F. Ablation: FCM vs. FreqSpatial

We compare against FCM (Feature Complementary Mapping) [11], originally proposed for aerial-image small-object detection in FBRT-YOLO. FCM performs channel-wise cross-attention via a 1:3 channel split with spatial and channel attention gates. It adds negligible parameters (20.33M, +0.15M) but achieves only 13.9% at epoch 0 (vs. 14.4% Baseline, 17.8% FreqSpatial) and reduces FPS to 48.2. Full 72-epoch FCM training is in progress; preliminary cold-start results indicate that frequency-domain processing, rather than generic attention, accounts for the improvement.

G. Qualitative Analysis

Fig. 3 shows representative detection comparisons. FreqSpatial detects objects missed by the Baseline in cluttered scenes and partially occluded cases. Objects with ambiguous spatial boundaries, where the Baseline fails to produce proposals, are recovered by FreqSpatial, consistent with the frequency branch’s global receptive field. Gains (Baseline $\rightarrow$ FreqSpatial) are highlighted in green.

H. Limitations

Several aspects of this study warrant further investigation and constrain the conclusions that can be drawn. First, we compared FreqSpatial only against the standard CSPRepLayer baseline; a parameter-matched or FLOPs-matched variant (e.g., a wider CSPRepLayer or deeper RepConv stack) would more precisely isolate the contribution of frequency-domain processing from that of increased model capacity. Second, all experiments were conducted on a single dataset (COCO 2017) and a single detector architecture (RT-DETRv2-S); cross-architecture and cross-dataset validation would test the generality of the frequency-aware approach. Third, results are reported from single training runs without multi-seed statistics; run-to-run variance should be quantified to confirm the reliability of the reported gains. Fourth, no component-level ablation (e.g., Scharr-only, FFT-only, per-FPN-level breakdown) has been conducted to isolate the contribution of each branch. Finally, the qualitative comparison is limited to two representative examples and would benefit from systematic per-category or per-occlusion-level analysis.

V. CONCLUSION

We introduced FreqSpatial, a frequency-spatial feature enhancement block that replaces the CSPRepLayer in RT-DETR’s FPN neck without backbone or decoder changes. By combining Scharr edge detection with FFT-based spectral convolution, FreqSpatial injects complementary frequency-domain information into multi-scale feature fusion.

On COCO 2017, FreqSpatial improves mAP by +1.45 (46.73% $\rightarrow$ 48.18%), reaching the official 120-epoch RT-DETRv2-S performance in only 72 epochs. Consistent gains

across object scales (+1.0–1.6 AP) and strong cold-start performance (+3.4 mAP at epoch 0) indicate that frequency-aware inductive bias benefits general-purpose object detection.

The main trade-off is the increased computational cost from FFT operations ($3.9\times$ FLOPs, $1.7\times$ latency increase), and the evidence is currently limited to one detector architecture and one dataset (Section IV-H). Future work should pursue more efficient frequency-domain operators, cross-architecture validation, adaptive frequency band selection, and extension to instance segmentation and panoptic tasks.

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Fig. 3: Qualitative detection comparison on COCO 2017 val. Each sample shows (left) Ground Truth, (middle) Baseline (46.73%), (right) FreqSpatial (48.18%). Green boxes denote objects detected by FreqSpatial missed by Baseline.